

EN 509: Functional Materials for Energy Engineering

Course Code	EN 509
Course Name	Functional Materials for Energy Engineering
L-T-P-C	3-0-0-3
Prerequisites	None
Intended for	UG/PG
Distribution	Compulsory for M. Tech. in Energy Engineering with specialization in Materials, and Elective for others
Mutual Exclusion	ME 609
Approval	9th Senate (May 2016)
Revised	11th Senate (Oct 2016) - revised

Course Contents

- **Introduction**

Definition of functional materials, Different kind of functional materials; Inorganic and organic functional materials, Metal organic framework based materials, Hybrid organic-inorganic Perovskites, Use of functionalities of materials in fabricating devices, Causes for observed functionality in a material; Functionality arising due to (i) electronic, (ii) spin, and (iii) ionic degrees of freedom; Exploitation of combined effects in designing new functional materials. [4 Lectures]

- **Functionality driven by electronic degrees of freedom**

Atoms and crystalline solids; electronic states of atoms and crystalline solids; Formation of bands in crystalline solids; Band dispersions; Density of states; Metals, semiconductors and insulators; Direct and indirect band gap semiconductors; Formation of impurity bands in the p-type and n-type semiconductors; Electrons effective mass in a semiconductor; Transport and optical properties of a semiconductor; Opto-electronic materials. [12 Lectures]

- **Functionality driven by spin degrees of freedom**

Formation of magnetic moment in an atom; Spin and orbital part of magnetic moment in a solid; Magnetization of a solid; Diamagnetic, paramagnetic, ferromagnetic, and antiferromagnetic materials; Different kind of antiferromagnetic structures; Exchange interaction; Determination of magnetic transition temperature using mean-field theory; Formation of domain wall in ferromagnetic material; Soft and hard ferromagnets; CMR/GMR materials. [10 Lectures]

- **Functionality driven by ionic degrees of freedom**

Covalent, ionic and metallic solids; Formation of dipole moment; Polarization of a material; Paraelectric, ferroelectric, antiferroelectric, piezoelectric, and pyroelectric materials; formation of domain wall in ferroelectric material; Multiferroic materials. [6 Lectures]

- **Functionality driven Energy Devices**

Energy efficient devices, Light emitting diodes, Power Electronic Devices, Quantum computers and devices, Opto-electronic devices, Thermoelectric Devices, Electro caloric and Magneto caloric devices, Photovoltaic Devices. [10 Lectures]

Text Books

1. N.W.Ashcroft and N.D. Mermin, **Solid State Physics**, Harcourt College Publishers, 1976.
2. Marius Grundmann, **The Physics of Semiconductors: An Introduction Including Devices and Nanophysics**, Springer, 2010.

References

1. R.M.Martin, **Electronic Structure: Basic Theory and Practical Methods**, Cambridge University Press, 2004.
2. K.F. Wang, J. – M. Liu, and Z.F.Ren, **Multiferroicity: the coupling between magnetic and polarization orders**, *Advances in Physics*, 58, 321, 2009.